

Chapter 7

Introductory quantum mechanics

7.1 Particle–wave duality

Quantum mechanical entities have both particle-like properties (position, discrete blobs of energy and momentum) and wave-like properties (frequency, wavenumber/wavelength, superposition). There is a direct correspondence between energy and momentum on the one hand, and frequency and wavenumber on the other, given by the *Planck–Einstein relations*,

$$E = h\nu = \hbar\omega, \quad p = \hbar k = \frac{h}{\lambda}, \quad (7.1)$$

where h is *Planck’s constant*,

$$h \equiv 6.62607015 \cdot 10^{-34} \text{ J s}, \quad \hbar \equiv \frac{h}{2\pi}. \quad (7.2)$$

Since 2018, the value of h given in (7.2) is *exact*, since it is taken to define the kilogram in the SI system (after the second is defined from the frequency of a certain radiation from caesium-133 and the metre is defined from the speed of light).

These relations also imply that a “particle” with position distribution $f(x)$ can be described as having wave number and hence momentum distribution

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{ikx} dx = \int_{-\infty}^{\infty} f(x) e^{-ipx/\hbar} dx = \tilde{f}(p/\hbar), \quad (7.3)$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{-ikx} dk = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \tilde{f}(p/\hbar) e^{-ipx/\hbar} dp. \quad (7.4)$$

The bandwidth theoreme relating the width Δx of $f(x)$ and the width Δk of $\tilde{f}(k)$,

$$\Delta x \Delta k \geq \frac{1}{2}, \quad (7.5)$$

translates into Heisenberg’s uncertainty (or indeterminacy) relation,

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (7.6)$$

Specifically, a particle with a precise location $x = x_0$ [so that $f(x) = 0$ if $x \neq x_0$] will have all possible momenta $-\infty < p < \infty$:

$$\tilde{f}(p) \propto e^{ipx_0/\hbar} \implies |\tilde{f}(p)| = \text{constant}. \quad (7.7)$$

Conversely, a plane wave $f(x) = Ae^{ipx/\hbar}$ has infinite extension and just a single definite momentum p .

7.2 Wave functions

All information about a quantum mechanical (non-relativistic) particle can be encoded in a complex function $\psi(x)$ called *the wavefunction*. This function corresponds to the probability of finding the particle in position x . More precisely, $|\psi(x)|^2$ is the *probability density*, ie

$$P(x_{\text{obs}} \in [x, x + dx]) = |\psi(x)|^2 dx \implies P(x_{\text{obs}} \in [x_1, x_2]) = \int_{x_1}^{x_2} |\psi(x)|^2 dx. \quad (7.8)$$

The probability of finding the particle anywhere between $-\infty$ and ∞ must be equal to 1, which gives us the *normalisation condition* (in 1 dimension),

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1. \quad (7.9)$$

In 3 dimensions the condition becomes

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\psi(x, y, z)|^2 dx dy dz = 1. \quad (7.10)$$

We could equally well describe the particle by a function of momentum,

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{ipx/\hbar} dx. \quad (7.11)$$

Then $|\tilde{\psi}(p)|^2$ is the probability density for momentum, ie,

$$P(p_{\text{obs}} \in [p, p + dp]) = |\tilde{\psi}(p)|^2 dp \quad (7.12)$$

7.3 The Schrödinger equation

The wavefunction $\Psi(\vec{r}, t)$ is a solution of the *Schrödinger equation*,

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi + V(\vec{r}) \Psi = i\hbar \frac{\partial \Psi}{\partial t}. \quad (7.13)$$

This equation can be “obtained” from the expression for the energy,

$$E = \frac{\vec{p}^2}{2m} + V(\vec{r}), \quad (7.14)$$

with the replacements

$$E \rightarrow i\hbar \frac{\partial}{\partial t}; \quad \vec{p} \rightarrow -i\hbar \nabla. \quad (7.15)$$

We now look for possible solutions of this equation of the form $\Psi(\vec{r}, t) = \psi(\vec{r})g(t)$. This gives us

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi(\vec{r})g(t) + V(\vec{r})\psi(\vec{r})g(t) = i\hbar \psi(\vec{r}) \frac{dg(t)}{dt}. \quad (7.16)$$

This can only work if we have

$$\frac{dg(t)}{dt} = Cg(t) \quad (7.17)$$

and

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r}) = K\psi(\vec{r}), \quad (7.18)$$

where C and K are constants. Inserting (7.18) into (7.16) we get

$$i\hbar \psi(\vec{r})g'(t) = K\psi(\vec{r})g(t) \quad (7.19)$$

$$\implies g(t) = Ae^{-iKt/\hbar} = Ae^{-i\omega t}. \quad (7.20)$$

We see that the constant $K = \hbar\omega = E$, and hence

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r}) = E\psi(\vec{r}) \quad (7.21)$$

The time-independent Schrödinger equation

Note that assuming

$$\Psi(\vec{r}, t) = \psi(\vec{r})e^{-i\omega t} = \psi(\vec{r})e^{-iEt/\hbar} \quad (7.22)$$

ensures that

1. total energy E is conserved, and
2. probability is conserved, since

$$|\Psi(\vec{r}, t)|^2 = |\psi(\vec{r})|^2 |e^{-i\omega t}|^2 = |\psi(\vec{r})|^2. \quad (7.23)$$

7.3.1 Superposition

We can easily see that any linear combination (superposition) of two solutions of the Schrödinger equation is also a solution,¹

$$\Psi(\vec{r}, t) = A_1\psi_1(\vec{r})e^{-iE_1t/\hbar} + A_2\psi_2(\vec{r})e^{-iE_2t/\hbar}; \quad A_1, A_2 \in \mathbb{C}. \quad (7.24)$$

¹To conserve probability we must also have $|A_1|^2 + |A_2|^2 = 1$.

This means that a quantum mechanical system or particle can be in a superposition of states with different energies. However, *only* the energies E_1 or E_2 will be measured in the case above, with

$$P(E_1) = |A_1|^2, \quad P(E_2) = |A_2|^2. \quad (7.25)$$

7.3.2 Free particle solution

The simplest case we can think of is $V(\vec{r}) = 0$, corresponding to a free particle. In this case the time-independent Schrödinger equation (7.21) is

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi(\vec{r}) = E\psi(\vec{r}) \quad (7.26)$$

In the one-dimensional case we have

$$-\frac{\hbar^2}{2m}\psi''(x) = E\psi(x) \quad (7.27)$$

$$\iff \psi''(x) = -\frac{2mE}{\hbar^2}\psi(x), \quad (7.28)$$

which we know has the solution

$$\psi(x) = Ae^{i(kx+\phi)}; \quad k^2 = \frac{2mE}{\hbar^2} \iff \hbar k = p = \sqrt{2mE}. \quad (7.29)$$

The last equation is exactly the relation $E = \frac{p^2}{2m} = \frac{1}{2}mv^2$.

The full (time-dependent) solution is

$$\Psi(x, t) = Ae^{i(kx-\omega t+\phi)} = Ae^{i(px-Et)/\hbar}e^{i\phi}. \quad (7.30)$$

The factor $e^{i\phi}$ is completely irrelevant since physics is completely unchanged by taking

$$\Psi \rightarrow \Psi' = e^{i\phi}\Psi, \quad (7.31)$$

since²

$$|\Psi'(x, t)|^2 = |e^{i\phi}|^2|\Psi(x, t)|^2 = |\Psi(x, t)|^2. \quad (7.32)$$

In three dimensions we get

$$\psi(\vec{r}) = Ae^{i\vec{k}\cdot\vec{r}} = Ae^{i\vec{p}\cdot\vec{r}/\hbar} \quad \text{with} \quad \frac{\vec{p}^2}{2m} = E \quad (7.33)$$

$$\Psi(\vec{r}, t) = Ae^{i(\vec{p}\cdot\vec{r}-Et)/\hbar} \quad (7.34)$$

— so a free particle is a plane wave!³

²This is not quite enough to prove that it is irrelevant. It is also required that all expressions of the form $\Psi^*(x, t)D\Psi(x, t)$, where D stands for some derivative, are also unchanged. It is easily seen that this condition is also satisfied.

³There is a problem with this, which is that there is no A such that $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$: plane waves are not normalisable. The solution to this problem is to construct a wave packet with finite size as a superposition of waves with (slightly) different momenta and energies.

7.3.3 Particle in a box

The next simplest potential is

$$V(x) = \begin{cases} \infty; & x < 0, \\ 0; & 0 \leq x \leq L, \\ \infty; & x > L. \end{cases} \quad (7.35)$$

This means that we must have $\psi(x) = 0$ for $x < 0$ and $x > L$, otherwise we would get $V(x)\psi(x) = \infty$ and E could not be finite. $\psi(x)$ must also be continuous everywhere, so we end up with the equation

$$-\frac{\hbar^2}{2m}\psi''(x) = E\psi(x) \quad ; 0 \leq x \leq L, \quad (7.36)$$

with boundary conditions

$$\psi(0) = \psi(L) = 0. \quad (7.37)$$

The solution is as before of the form

$$\psi(x) = A_+e^{ikx} + A_-e^{-ikx}, \quad k = \sqrt{\frac{2mE}{\hbar^2}}. \quad (7.38)$$

$$\psi(0) = A_+ + A_- = 0 \quad \Rightarrow A_- = -A_+ = -A. \quad (7.39)$$

$$\psi(L) = A(e^{ikL} - e^{-ikL}) = 2iA \sin(kL) = 0 \quad (7.40)$$

$$\Rightarrow kL = n\pi \quad \Rightarrow k_n = n\frac{\pi}{L} \quad (7.41)$$

$$\Rightarrow E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2}{2mL^2} n^2. \quad (7.42)$$

We see that only discrete energy values E_n are allowed.

We can also find the constant A :

$$\begin{aligned} \int_{-\infty}^{\infty} |\psi_n(x)|^2 dx &= |A_n|^2 \int_0^L 4 \sin^2 \frac{n\pi x}{L} dx = 4|A_n|^2 \int_0^L \frac{1}{2} \left[1 - \cos \frac{2n\pi x}{L} \right] dx \\ &= 2|A_n|^2 \cdot L = 1 \end{aligned} \quad (7.43)$$

$$\Rightarrow A_n = \frac{e^{i\phi}}{\sqrt{2L}}, \quad (7.44)$$

where ϕ is an arbitrary phase (which we usually set to 0).

7.3.4 Finite potential: tunneling

Now consider a potential

$$V(x) = \begin{cases} 0; & x < 0 \\ V; & x \geq 0. \end{cases} \quad (7.45)$$

The solution will be a solution to the equation $H\psi(x) = E\psi(x)$ in both regions, with the *same* E everywhere.

For $x < 0$ we can write as before,

$$\psi_{<}(x) = A_+e^{ikx} + A_-e^{-ikx}. \quad (7.46)$$

For $x > 0$ we write

$$\psi_{>}(x) = B_+ e^{qx} + A_- e^{-qx}. \quad (7.47)$$

It can be show that all values for $E > 0$ are possible in this setup. We can distinguish between three cases

1. $E > V$

$$-\frac{\hbar^2}{2m}\psi''(x) + V\psi(x) = E\psi(x), \quad x \geq 0 \quad (7.48)$$

$$\iff \psi''(x) = -\frac{2m(E - V)}{\hbar^2}\psi(x) \quad (7.49)$$

$$\iff \psi(x) = B_+ e^{ik'x} + B_- e^{-ik'x} \quad \text{with} \quad \hbar k' = \sqrt{2m(E - V)}. \quad (7.50)$$

The $+$ solution represents a right-moving wave, while the $-$ solution represents a left-moving wave. If we assume the only “source” is at “ $x = -\infty$ ” (ie, a particle coming in from the left) we can set $B_- = 0$. Both the wavefunction and its derivative must be continuous at $x = 0$, so matching the solutions on either side we get

$$\psi(0) = A_+ e^{ik \cdot 0} + A_- e^{-ik \cdot 0} = B_+ e^{ik' \cdot 0} \quad (7.51)$$

$$\iff A_+ + A_- = B_+ \quad (7.52)$$

$$\psi'(0) = ikA_+ - ikA_- = ik'B_+ \quad (7.53)$$

$$\iff \frac{k}{k'}(A_+ - A_-) = B_+ = A_+ + A_- \quad (7.54)$$

So we find

$$A_- = \frac{\frac{k}{k'} - 1}{\frac{k}{k'} + 1} A_+ = \frac{k - k'}{k + k'} A_+, \quad (7.55)$$

$$B_+ = A_+ + A_- = \frac{2k}{k + k'} A_+. \quad (7.56)$$

$A_+ e^{ikx}$ is the incoming (*incident*) wave

$A_- e^{-ikx}$ is the *reflected* wave

$B_+ e^{ikx}$ is the *reflected* wave.

2. $E = V$

$$-\frac{\hbar^2}{2m}\psi''(x) = (E - V)\psi(x) = 0, \quad x \geq 0 \quad (7.57)$$

$$\implies \psi(x) = B_1 + B_2 x. \quad (7.58)$$

However, the second term is unphysical since it diverges (goes to ∞) as $x \rightarrow \infty$. Therefore, in this limiting case we get the $\psi(x) = B_1 = \text{constant}$ for $x \geq 0$.

3. $E < V$

$$\psi''(x) = \frac{2m(V - E)}{\hbar^2}\psi(x) \quad (7.59)$$

$$\implies \psi(x) = B_+ e^{rx} + B_- e^{-rx} \quad \text{with} \quad \hbar r = \sqrt{2m(V - E)}. \quad (7.60)$$

The B_+ term is unphysical since it diverges as $x \rightarrow \infty$. The B_- term describes *quantum tunneling* – even if $E < V$ there is still a finite probability of finding the particle at $x > 0$.

Considering the matching conditions again, we find

$$\psi(0) = A_+ + A_- = B_- \quad (7.61)$$

$$\psi'(0) = ikA_+ - ikA_- = -rB_- \quad (7.62)$$

$$\implies -r(A_+ + A_-) = ik(A_+ - A_-), \quad (7.63)$$

which gives us

$$A_- = \frac{ik + r}{ik - r} A_+ = \frac{k - ir}{k + ir} A_+, \quad (7.64)$$

$$B_- = \frac{2k}{k + ir} A_+. \quad (7.65)$$

Note that all the coefficients A_+, A_-, B_- are complex. We can *choose* an overall phase so that B_- is real and $A_- = A_+^*$:

$$A_+ = Ae^{i\phi} \quad \text{with} \quad \tan \phi = \frac{r}{k} \quad (7.66)$$

$$\implies A_- = Ae^{-i\phi}, \quad (7.67)$$

$$B_- = \frac{2kA}{\sqrt{k^2 + r^2}}. \quad (7.68)$$